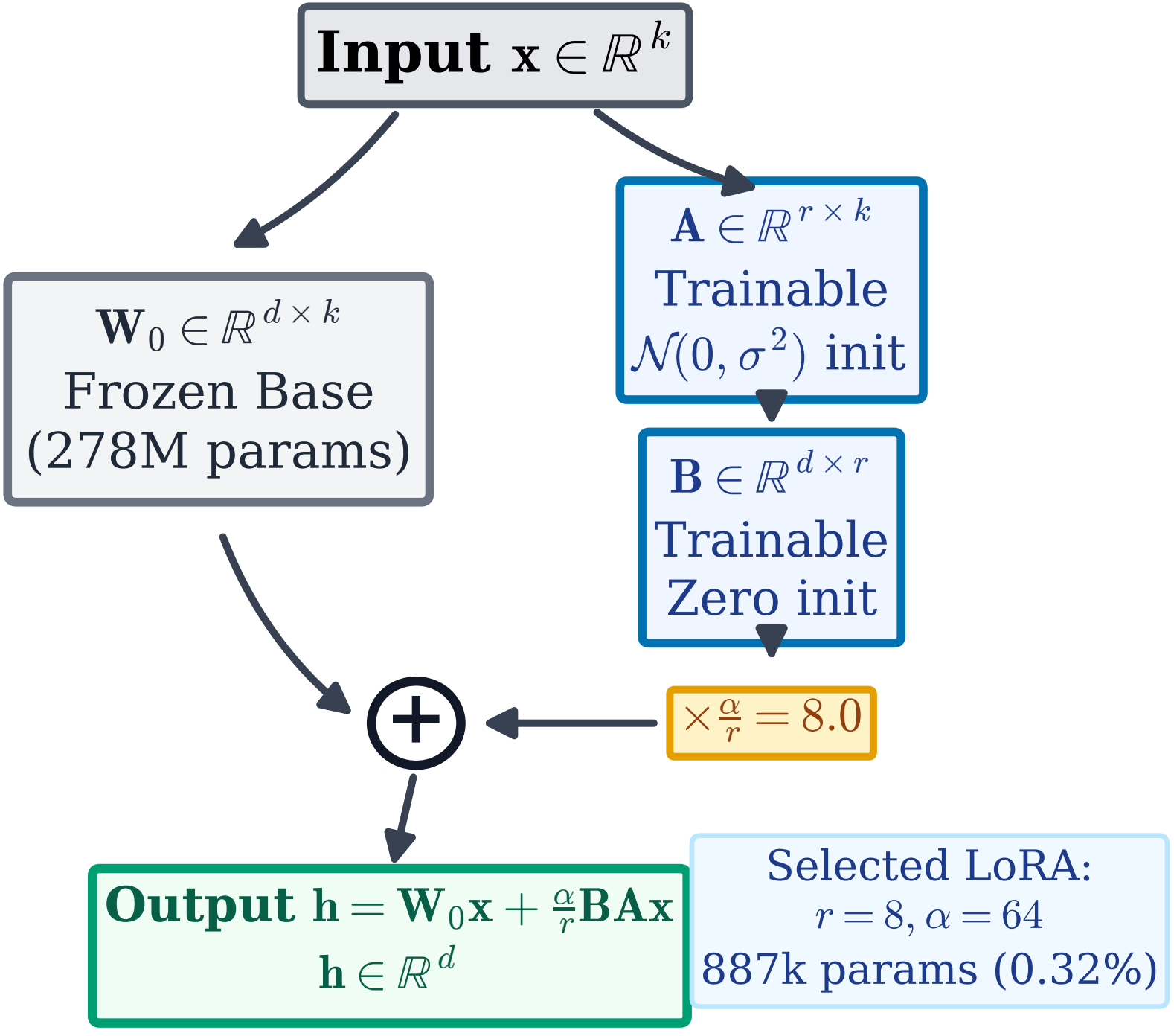
Input $\mathbf{x} \in \mathbb{R}^k$
$\mathbf{W}_0 \in \mathbb{R}^{d \times k}$
Frozen Base
(278M params)
$\mathbf{A} \in \mathbb{R}^{r \times k}$
Trainable
$\mathcal{N}(0, \sigma^2)$ init
$\mathbf{B} \in \mathbb{R}^{d \times r}$
Trainable
Zero init
$\times \frac{\alpha}{r} = 8.0$
Output $\mathbf{h} = \mathbf{W}_0\mathbf{x} + \frac{\alpha}{r}\mathbf{BAx}$
$\mathbf{h} \in \mathbb{R}^d$
Selected LoRA:
$r = 8, \alpha = 64$
887k params (0.32%)